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A Secure radio communication system based on an efficient speech watermarking approach

R. Fantacci, S. Menci*,†, L. Micciullo and L. Pierucci

Department of Electronics and Telecommunications, University of Florence, Via di S. Marta 3, 50139 Florence, Italy

Summary

This paper deals with an efficient secure communication system based on a speech watermarking approach. The main goal of the system is to efficiently support air traffic control (ATC) communications by improving safety and security of the existing ATC equipment embedding digital side information into speech signals in order to allow an *automatic identification of the talker*. An implementation based on the use of the well known LPC (*linear predictive coding*) principle is discussed. The LPC residual is split up into voiced and unvoiced segments; while the voiced segments cross the system unmodified, the watermarking data are embedded into the unvoiced segments, spectrally similar to a white noise. The watermark signal is decoded by a very similar scheme. We provide also a statistical analysis on the unvoiced parts of the ATC communications that demonstrates the system feasibility. In addition to this, an optimization of the entire system has been also performed to increase performance. The obtained results highlight that the watermark data are nearly imperceptible and provide *a data channel with rate from 2000 to 5000 kbits/s*, with *high robustness to channel noise*; the system requires only modular, low-cost and easy-to-install encoders and decoders, maintaining compatibility with legacy RF DSB-AM transceivers. Moreover, it is well suited both for the actual 25 kHz RF channels and for the new 8.33 kHz European RF channels. Copyright © 2008 John Wiley & Sons, Ltd.

KEY WORDS: secure communications; watermarking; speech processing; air traffic control; LPC; voiced/unvoiced segmentation

1. Introduction

Secure communications are nowadays a main concern in different communication systems including wired, wireless and optical transmission platforms. Security procedures are usually adopted to protect the traffic confidentiality, integrity and to prevent different types of network security attacks [1]. In particular, after the 11 September terroristic attack, these requirements have become of paramount importance for ATC (*air traffic control*) communications.

Almost all tactical ATC communications are handled using DSB-AM radio systems. DSB-AM voice channels are allocated sector-wise and their

*Correspondence to: Simone Menci, LaRT (Laboratory of Networks and Telematics), Department of Electronics and Telecommunications, University of Florence, Via di S. Marta 3, 50139 Florence, Italy.
†E-mail: simone.menci@unifi.it

deployment requires careful frequency planning. The operation of voice communications between an air traffic controller and all pilots of aircrafts (ACs) within a controlled flight sector is on the same VHF (Very High Frequency) working frequency (party line communications) and is completely manual.

To identify a speaking AC, pilots have to start all verbal communications with the AC call-sign. However, 'mis-identification' (*call-sign confusion*) is most likely to happen due to low quality of the VHF transmission, to similar call-signs of different ACs, to errors of comprehension/speaking and so on. The correct identification of who is talking and with whom is very important not only to improve safety but also to prevent fake AC VHF messages from jokers, terrorists, or unauthorized third parties. *This security lag can be overcome if an automatic, hidden sender identification and displaying system can be provided without the necessity of human interaction.*

Digital Watermarking techniques may be a possible solution [2–6]. Watermark was applied first on paper and nowadays on pictures, digital videos, audio sources for copyright and IPR (intellectual property right) protection. Watermark information is a special kind of bit pattern that behaves as an intentional noise inserted into the digital audio/image/video signal to identify the copyright information such as author, recording label, and usage rules. In addition, it can be used to resolve rightful ownership and it can provide evidence of copyright infringements after the copyright violation has occurred. Watermarking algorithm must satisfy at least two constraints: imperceptibility (or inaudibility) and robustness to withstand attempts such as removal or alteration of inserted watermarks.

In ATC secure communications watermark technique was adapted to the low transmission quality: noisy channels and narrow signal bandwidth constrain the additional hidden data embedded into the speech signal compared to the requirements of CD's watermarking. Among different alternatives for speech watermarking, as direct sequence spread spectrum (DS-SS) [7], quantization index modulation (QIM) [8,9], frequency masking [10], the paper uses the approach of DS-SS applied only in the unvoiced part of the speech signal [11], combined with two key technologies, *LPC* (*linear predictive coding*) *analysis/synthesis* and *voiced/unvoiced segmentation*. Given a speech signal, the coefficients of the vocal tract filter are adaptively obtained by a linear prediction based on previous samples. Direct FIR (Finite Impulse Response) filtering of the speech signal results in the so-called prediction residual. The LPC residual is split up into voiced and unvoiced segments; while the voiced segments pass the system unmodified, the watermarking data are embedded into the unvoiced segments that have a behavior like noise. In this way, the code added to the speech signal should not degradate significantly the speech signal quality.

Unfortunately, the unvoiced part of speech is not continuous, making the data transmission impulsive. A goal of this paper is to provide a statistical analysis on the voiced and unvoiced parts of the VHF air/ground voice communications that demonstrates the feasibility and the effectiveness of this watermarking approach for the AC signature identification service and for new aeronautical services envisioned in the future. An optimization of the entire system and in particular of the segmentation algorithm has been performed. This activity suggests also a simple solution for making this system adaptive in terms of a trade-off between data rate and speech quality. Simulation results on a realistic VHF channel model show that performance requirements can be met with an adequate data protection technique.

The watermark system improves safety and security of the existing ATC equipment, requiring only low-cost and easy-to-install encoders and decoders, and maintaining compatibility with legacy RF DSB-AM transceivers. Recently, new 8.33 kHz voice RF channels were introduced in the upper flight sectors of the European flight area in order to efficiently face the requests of new channels originated from the fast growth of the air traffic. The proposed watermarking system is well suited both for the actual 25 kHz RF channels and for the new 8.33 kHz European RF channels.

This paper is organized as follows. Section 2 presents a brief review of the current state of the art of speech watermarking, with particular focus on ATC applications and on constraints and drawbacks of traditional approaches. Section 3 shows the main functional blocks of the watermarking system with description of each key operation with particular attention on LPC analysis/synthesis, voiced/unvoiced segmentation and watermark embedding techniques. Section 4 presents the feasibility analysis carried out in order to demonstrate that achievable data rate and transmission continuity meet the performance requirements for the AC identification tag service. Section 5 presents simulation results in terms of system optimization, segmentation correctness, data and speech quality. Finally, the conclusions (Section 6)

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highlight the advantages and open issues of this approach focusing on the novel results obtained in this work.

2. Speech Watermarking: A Brief Review

Digital watermarking is based on the embedding of digital data into video, image or audio signals without degrading the perceptual quality of the host signal. In the last decade, watermarking has received considerable attention from various application fields including traitor tracing, authentication, copy prevention, broadcast monitoring, steganography, archiving and legacy system enhancement. System designs vary depending on the type of host signal (video, image, or audio); our focus lies on *speech watermarking* (SW) for transmission of additional side information over the legacy voice radio communication link between AC and ATC center [12–16]. This implies *blind watermarking*, where the decoder does not know the original host signal, and transmissions over a VHF channel.

At the beginning, watermarking was intended as a purely *additive* process (Figure 1a). It was soon pointed out that the data perceptibility could be reduced when the watermark signal was time or spectrally shaped. The principle is shown in Figure 1(b): the data undergoes (usually adaptive) filtering and then is added to host signal. The prominent class of *spread-spectrum watermarking systems*, which are based on the detection of pseudorandom noise sequences, belongs to this category. Theoretical considerations, however, highlighted an intrinsic constraint on the achievable trade-off between data rate, speech quality, and robustness due to the interference between host signal and watermark. As a consequence, the watermark has to be kept, for perceptual reasons, usually 16–20 dB below the speech signal, making the watermark detection a difficult task. As showed in the current state-of-the-art ATC SW systems ([12,13,15]), bit rates in the range 30–200 bits/s can be achieved.

Additive embedding can occur as well in a *transform domain* of the signal (Figure 1c). This is commonly done in other watermarking applications (e.g., image processing) but is not commonly adopted for speech watermarking. A different watermarking approach [2] is based on the concept of modifying the host signal or its transform domain representation, instead of adding the watermark data signal. QIM [8,9] is an example (Figure 1d); various enhancements of this technique with bit rates in the range from 3 to 300 bits/s have been proposed. In a generalization of QIM the signal or one of its parameters is *modulated* by the watermark (Figure 1e); this type of methods achieve bit rates in the range of 400 bits/s.

It can be concluded that the reported bit rates vary widely, from few bits/s up to 400 bits/s. This is mostly due to the fact that great care is taken to maintain the original shape or properties of the speech signal in order to minimize perceptual distortion. In our experience [15], data reliability considerations make difficult to overcome the threshold of 100 bits/s in practical applications; hence *this rate value will be considered, throughout this paper, as the practical maximum data rate for actual speech watermarking systems based on traditional approaches*.

A different approach is presented in Reference [17]. In segments where some frequency components of an audio signal above 5 kHz have a noise-like structure, they can be replaced by a pseudorandom signal, modulated by the watermark (Figure 1g) without introduce distorsion. This method is not directly applicable to low bandwidth radio speech. The method showed in this paper uses a similar approach for speech signals, based on how the speech is produced and perceived and on the working principles of LPC vocoders.

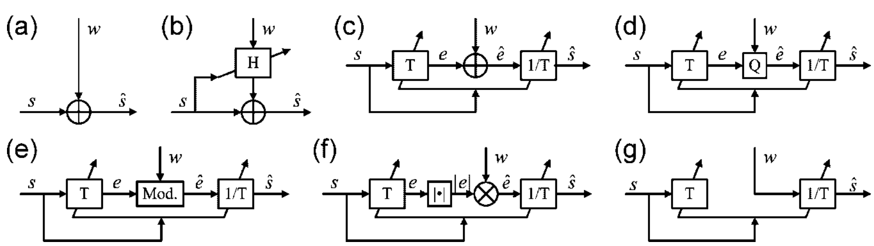

Fig. 1. A watermarked signal $\hat{s}$ can be generated from the original speech signal s and the data signal d both adding d to s directly (a) or after an adaptive filtering (b), and applying the data onto a transform domain signal representation e through addition (c), quantization according to the data (d), modulation (e), and, as showed in this paper, multiplication of $|e|$ (f) and replacement of e (g) by the data.

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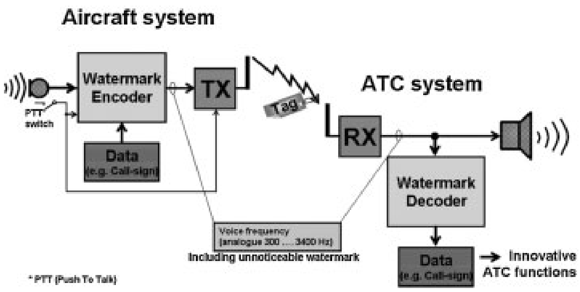

Fig. 2. Functional scheme of a generic aeronautical Speech Watermarking system.

3. Watermarking Scheme

Figure 2 shows the functional scheme of a classical ATC Speech Watermarking system [18]. The watermark encoder embeds the addresser identification as a small digital tag into the voice signal. It is therefore inherently transmitted with the voice signal; at the receiving end, the addresser identification can be detected by a watermark decoder.

LPC of speech signals was first presented by Atal and Schroeder [19] almost 40 years ago and found its use into many prominent low-rate speech coders [20]. Figure 3 shows the underlying source filter model. Given a speech signal, the coefficients of the vocal tract filter are obtained by a linear prediction based on previous samples. Direct (FIR) adaptive filtering of the speech signal results in the so-called prediction *residual* (or voice *excitation*) e.

It has been shown in the context of speech vocoders that the source filter model provides a very accurate representation for the *unvoiced* components of speech. A listening test with 32 subjects [22] showed that resynthesized speech with pure white noise excitation in the unvoiced segments leads to MOS (*mean opinion score*)[‡] that are almost identical to those of unmodified PCM (*pulse code modulation*) speech. The conclusion is that *the speech quality does not degrade when the LPC residual in unvoiced segments is replaced by white noise of equal power.*

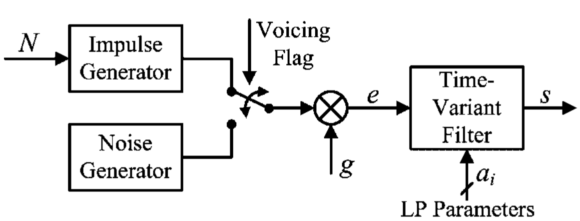

Fig. 3. Source filter model of speech production [21]. For voiced segments the excitation e created by the vocal chords is a periodic pulse train with a pitch period N, and a random noise signal for unvoiced segments. Both are amplified by a time-variant gain g. A parametric all-pole filter models the vocal tract.

The adopted watermarking system, which belongs to the family of modulation models (Figure 1e), is based on this approach. The system can alternatively apply two rather simple types of modulations. The first is the *multiplication* of the speech signal in a transform domain with the data signal (Figure 1f). The second is the complete removal of the original transform domain signal and a *replacement* by the watermark signal as depicted in Figure 1. We selected the LPC residual of *unvoiced* speech segments as transform domain signal representation. The basic structure of the resulting watermarking system is shown in Figure 4 (embedder) and 4 (decoder).

3.1. LPC Analysis/Synthesis Framework

Speech analysis and resynthesis are based on the LPC vocoder approach as discussed in previous Section 3. The LPC residual is split up into voiced (V) and unvoiced (U) segments. The same LP filter coefficients are used for the analysis and for the inverse synthesis filter. The coefficients are estimated from the 8 kHz speech signal and periodically updated: a 10th-order LP analysis is updated every 30 samples using the *Levinson-Durbin* algorithm [25] and a window length of 160 samples. Each frame is divided into two sub-frames of 15 samples. The residual is computed for each sub-frame. After potential modification by the watermark embedding, the speech signal is resynthesized using the LPC parameters set.

3.2. Voiced/Unvoiced Segmentation

As a basis for the V/U segmentation, a pitch estimation is computed with the autocorrelation-based pitch tracking algorithm as implemented in PRAAT program [26]. This algorithm has five important parameters that may affect the segmentation: *Number of Candidates*, *Silence Threshold*, *Minimum Pitch*, *Ceiling* (Maximum Pitch), and *Voicing Threshold*. We initially used a

[‡]Mean opinion score is an evaluation scale commonly adopted for the subjective measurement of voice quality [23,24]; the scale goes from 1 (bad quality) to 5 (best quality), and is evaluated through extensive campaigns involving a sufficiently high number of listeners, each one giving his or her opinion on the audio quality.

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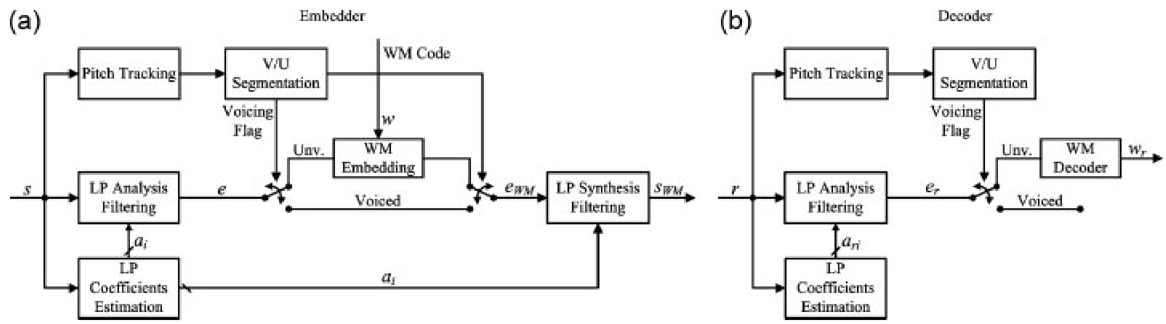

Fig. 4. Block diagrams of the high rate speech watermarking scheme: (a) embedder and (b) decoder.

fundamental frequency range of 60–500 Hz and a *Voicing Threshold* of 0.35, which is below PRAAT's default value of 0.45 and leads to more voiced and less unvoiced segments. This bias towards voiced components ensures that these perceptually important segments remain unmodified in the later processing steps. We found that the first two parameters do not affect significantly the segmentation, while the others are relevant for the system optimization; as we will see in Section 5.2, a better voice segmentation can lead to a higher quality for watermarked voice and a better robustness of segmentation to channel noise.

3.3. Watermark Embedding

The watermark data are assumed to be a binary coded random signal with values ± 1. We analyzed the following three methods of watermark embedding.

Using the *multiplication* method depicted in Figure 1, the absolute value of each sample of the residual is multiplied by a sample of the watermark signal. The power of the residual is thereby accurately maintained, and the information is encoded in the sign of the residual.

The same rate is achieved with the *replacement* method in which the residual is replaced by the binary watermark signal (Figure 1g). For each sub-frame, the power of the watermark signal is matched with the power of the original residual; mean power matching has given better results than instantaneous power matching.

The more robust *replacement* plus *spreading* method is based on the previous replacement concept. In each sub-frame the watermark bits are modulated by a pseudo-random noise (PN) sequence with fixed length. Again, the power of each sub-frame is matched with the power of the original residual.

3.4. Watermark Decoding

The receiver must be able to extract, from the received signal, the watermark code. The watermarked voice signal is sent directly to the output; this is possible since the watermark signal, thanks to the V/U-based integration process, has not significantly altered the voice quality. As a coexistence constraint, the receiver cannot perform a cancellation of data from the received voice because this signature transmission system must be transparent to the user; the voice should be received without problems also by legacy receivers (*imperceptibility constraint*).

The watermark decoder obeys the same structure and processing steps as the embedder (Figure 4b). For both the multiplication and the replacement methods, the sign of the blindly re-estimated residual of the received speech signal yields the detected watermark bit. In the spreading method, the sign of the maximum of the cross-correlation between a sub-frame of the residual and the PN sequence determines the watermark bit.

The decoder is very similar to the encoder: this almost perfect symmetry is an advantage as *it is possible to realize an hardware or software modular implementation, with a slight customization for transmitter or receiver, and lower design and deployment costs*.

4. System Feasibility Analysis

The main performance requirements of a such system are the following:

- *Audibility*: the data stream should not disturb the voice communication and still be reliably transmitted. This is important also for *compatibility with not watermarking-ready legacy RF facilities*. MOS should be ≥ 3.6.

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- *Data Rate*: the minimal required data rate is 40 bits/s[§]. Hundreds or thousands of bits/s are useful to deploy new ATC data services.
- *Reliability*: watermarking message bits should be detected with a *bit error rate* (BER) $\leq 10^{-3}$.
- *Delay*: Transmission/decoding delay should be kept < 1 s.
- *RF Channels*: the system must be adaptable to operate in both European RF channel bandwidths, 25 and 8.33 kHz.

This section deals with the feasibility of such a system. In particular, the main concern here is the capability of embedding a stream of watermark data sufficient to support the AC signature system, according to the above mentioned performance requirements. First, it is fundamental to evaluate the real mean value of the achievable bit rate. Moreover, differently from previous approaches [12,15], the data embedding is possible only when speech is in the unvoiced state, hence, allowing *data transmissions in a burst mode*.

As for the resulting mean data rate, evaluating many samples of speech signals registered from ATC radio communications, we found that the mean value of the percentage of unvoiced frames is in the range 30–34%. This rather high value can be considered due to frequent short silence pauses and well spoken words between the pilot and the ground controller. Assuming a sampling frequency of 8 kHz, we have, for the first two embedding methods (1 bit/sample), a feasible mean bit rate of 2.4 kbits/s, that is *more than 20 times higher than current systems*, which, as shown in Section 2, achieve about 100 bits/s with sufficient reliability. The spreading method yields a lower rate which depends from the selected spreading factor.

Coherently with the LPC analysis, frames of 30 samples are treated as transmission units (packets). With one or two packets we are able to identify a sufficient number of ACs/controllers, but we must assure, *as a service target, that it is always possible to transmit at least a packet per second*. So, we evaluated the statistics of the duration of voiced periods in order to understand what is the distribution of the pauses in data transmission. Figure 5 shows the empirical

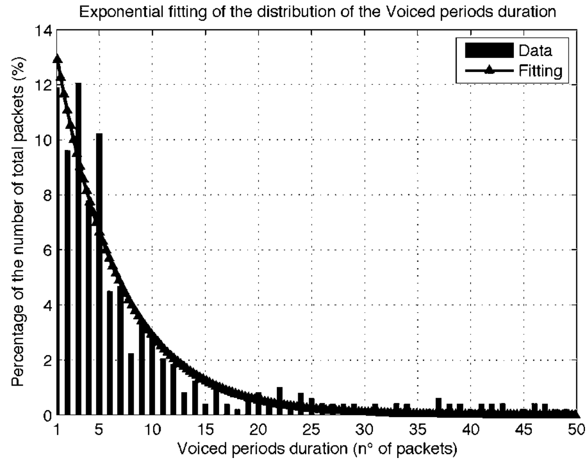

Fig. 5. Empirical distribution and exponential fitting of the voiced periods duration.

distribution obtained through simulations for many ATC communications. The trend is quasi-exponential (long continuous voiced periods have low occurrence); the exponential fitting is also shown in the figure[‖]. We evaluated, with a confidence of 99%, the expected value of Voiced periods maximum length, both from empirical distribution and analytical fitting:

$$\int_0^y p(x)\,\mathrm{d}x = 0.99, \qquad \text{where } p(x) = b \cdot e^{-b\cdot x}$$

and b defines the exponential distribution trend. Considering all audio samples (single or combined), the result is $y \leq 28$: the voiced periods do not last more than 28 packets, that is $28 \times 30/8000 = 0.105\text{ s} < 1\text{ s}$, meeting the requirement of a 'sufficiently continuous' transmission and demonstrating the system feasibility for the signature detection service.

5. Simulation Results

5.1. Work Hypotheses and Simplifying Assumptions

In order to evaluate the performance degradation due to the watermarking signal, we assume an ideal frame synchronization between encoder and decoder. This condition in practical applications can be achieved by resorting to conventional synchronization schemes

[§]A few tens of bits for each message are required in order to have a sufficient number of signatures for the European air traffic: 40 bits is set as the minimum target because assuming, as an example, 16 bits for ancillary fields (timestamp, CRC, and others), the other 24 bits can carry $2^{24} \approx 16$ million signatures, sufficient to satisfy current and future needs.

[‖]A more precise Pareto fitting has given similar results, both qualitatively and numerically.

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such as pilot sequence embedding [27]. The spreading sequence, when used in the watermark signal, can also be helpful in this receiver task.

For the development and simulation of this system we used MATLAB/Simulink R2007b environment. In deriving our simulation results we made the following assumptions:

- *Audio sources*: a group of proof signals, in `.wav` format, obtained recording ATC communications directly in the AC cockpit in order to not collect channel noise.
- *RF Channel*: direct link or VHF ATC channel [28]; thanks to the envelope-detector and low-pass filter in the RF section of the receiver, Doppler shift has no effect, and the fading effect is reduced, giving results which we found similar to a simple AWGN channel.
- *SNR*: evaluated at the decoder input, it has been varied from 20 to 50 dB, although low values in the range 20–30 dB are seldom encountered in the usual ATC operativity.
- The five parameters of the V/U segmentation algorithm (Section 3.2) have been varied in order to achieve the best performance (*optimization procedure*).
- *Spreading sequences* with lengths 5, 10, and 15 chips; as a consequence, each unvoiced frame carries 2, 3, or 6 bits.
- 10 000 s of speech and data transmitted per simulation run.

Performance of the proposed system have been evaluated in terms of BER for the quality of received watermark data. Evaluation of voice quality is not easy, since extensive and expensive subjective listenings are needed to make it feasible. Hence, we started our analysis by performing a limited listening campaign, associated with automatic measurement of MOS through ITU's PESQ-LQ (*perceptual evaluation of speech quality—listening quality*) method [29,30]. PESQ-LQ is a score homogeneous with subjective MOS scale, and is a very useful method for speech quality assessment. However, *we found PESQ-LQ useless for this type of watermarking*: since the comparison between original and watermarked signals is made in a perceptive domain, and this type of watermark is embedded on the vocal tract excitation, the method fails in providing a realistic measure of the difference of speech quality. Therefore, the classical MOS measurements for watermarked speech quality are used.

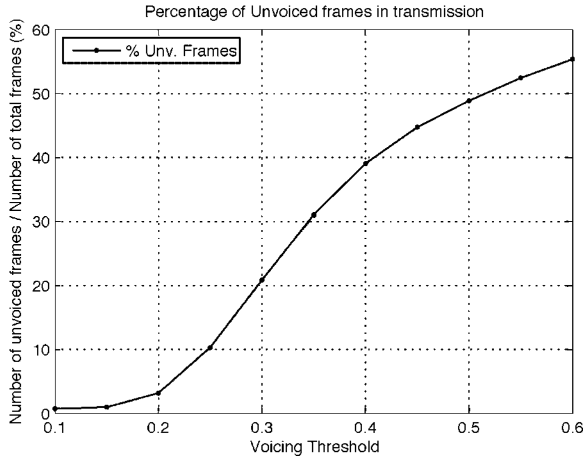

Fig. 6. Unvoiced frames in transmission as a function of *Voicing Threshold*.

5.2. System Optimization

The first set of simulation runs focused on the direct link with the goal of maximizing the speech quality, data reliability and system throughput by searching for the optimal parameters value of the pitch tracking algorithm. As results of our analysis we found that *Silence Threshold* and *Number of Candidates* have no impact on the performance, while optimal values for *Minimum Pitch* and *Ceiling* resulted to be 250 Hz and 950 Hz, respectively, for a MOS value close to 4.4 (i.e., imperceptible watermark). The influence of the *Voicing Threshold* parameter on the system performance is investigated in Figure 6. This figure shows that in the range from 0.1 to 0.6 the percentage of unvoiced segments grows accordingly with the threshold, which represents the 'sensitivity' of the algorithm to the strength of pitch candidates. Moreover, it is also evident in the figure that in the range from 0.2 to 0.45 the trend is almost linear; this suggests us *the use of this parameter to dynamically control the quantity of data 'injected' in the voice signal. This makes it possible to reach high data rates (up to 5000 bits/s)*, obviously at the expense of a voice quality and data reliability degradation (however a suitable trade-off between data rate and voice and data quality is always achievable).

5.3. Segmentation Results

After completing the optimal system parameters setting and assumed a *Voicing Threshold* value of 0.35 we investigated the influence of the ATC channel on the system performance. The segmentation is assumed performed differently at the embedder and the decoder

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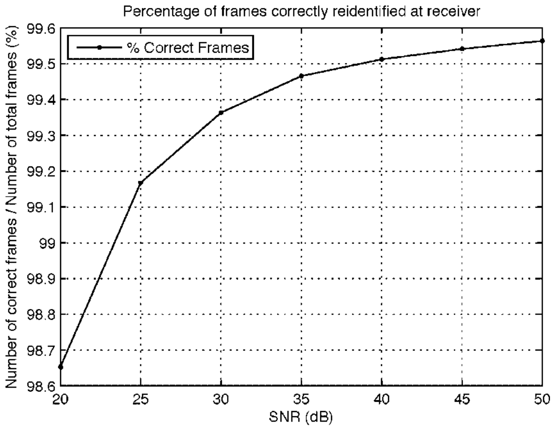

Fig. 7. Frames correctly reidentified in the decoder as a function of SNR.

side in order to take into account of the influence of the ATC channel and insertion of watermark data on the voice. Figure 7 shows that the percentage of correctly identified frames is very high and that it is slightly dependent on the SNR value (<1%). From above, it follows that *the segmentation process is robust (i.e., it has low sensitivity) with respect to the channel noise*. Figure 8 shows $(V)_{tx} \rightarrow (U)_{rx}$ and $(U)_{tx} \rightarrow (V)_{rx}$, that is the percentages of frames, respectively V/U in transmission (tx), not correctly reidentified at the receiver (rx): these errors are very low and slowly decreasing with SNR (white noise has a whitening ('unvoicing') effect on the received speech).

However, in a more advanced implementation, it should be mandatory a perfect segmentation in order to further minimize BER. A possible solution is the insertion, in each encoded watermark frame, of data useful to assist the decoder's V/U estimation (for example, a pointer to the next frame containing a watermark).

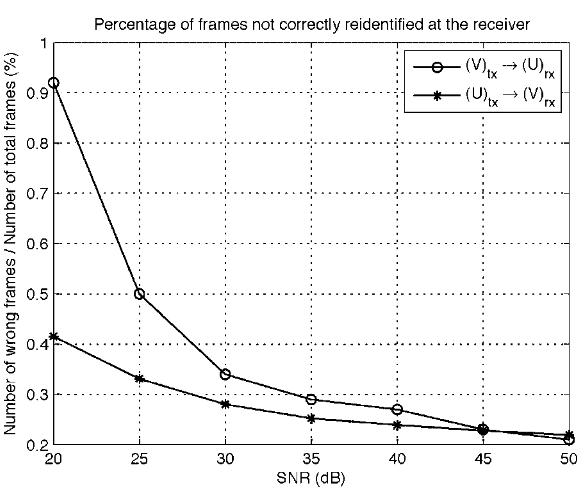

Fig. 8. Frames not correctly reidentified ($(V)_{tx} \rightarrow (U)_{rx}$ and $(U)_{tx} \rightarrow (V)_{rx}$) in the decoder as a function of SNR.

5.4. Received Data Quality

This section deals with the received data quality evaluation. In particular, the obtained results derived by computer simulations *have highlighted worse performance of the multiplication method with respect to the replacement method*. This behavior is due to the fact that while replacement method guarantees the same minimum power for all residual samples, in multiplication method residual values are not scaled so that samples with low amplitude have a high probability to give rise to bit errors. Consequently, in what follows we will focus mainly on the replacement method (Figure 9). Unfortunately, the obtained numerical results show that the service performance target of BER $\leq 10^{-3}$ is not met, even under high SNR values, where the performance floor (BER $= 0.015$) is due to the effect of the unavoidable interference between voice and data.

In order to overcome these residual errors, we have to introduce some form of data protection. As a rough approach to lower this drawback, we have considered here the integration of spreading codes with replacement embedding, as explained in Section 3.3. The obtained performance results are shown in Figure 10. It is evident from this figure that spreading codes with 10 chips makes the system able to meet the target BER constraint. Moreover, with a length of 15 chips the floor is completely eliminated. Despite those results, this approach does not represent an efficient solution for several applications due to the

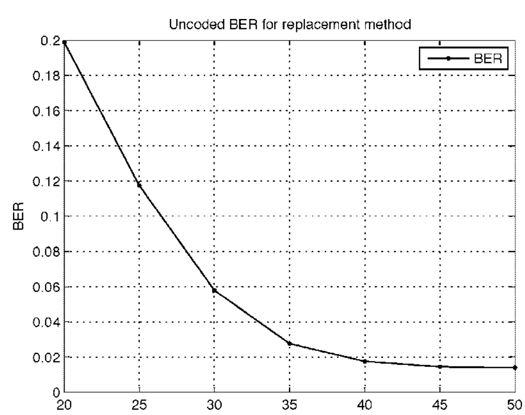

Fig. 9. Uncoded BER for replacement method as a function of SNR.

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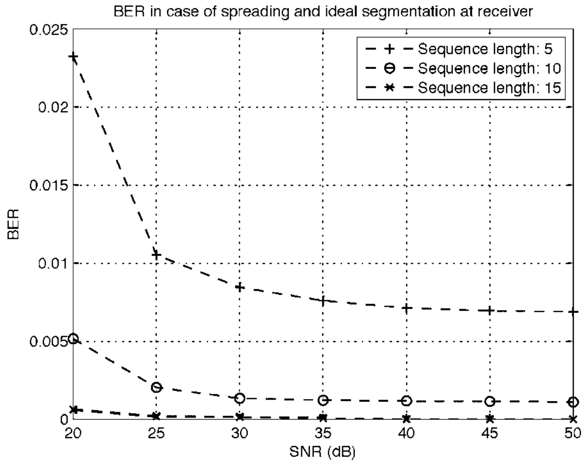

Fig. 10. BER for replacement plus spreading method as a function of SNR.

resulting data rate. As a consequence, the development of a more efficient data protection technique is mandatory if we want to fully exploit the capacity of this hidden data channel.

5.5. Perceptual Quality of Received Speech

From the obtained results, we had that the estimated quality of received voice signal is very good (MOS $\geq$ 4.4), higher than previous SW systems for applications in ATC systems as those proposed in References [15,16]. Only for very low SNR values or in the case of a non-optimized segmentation algorithm, careful subjective listening over headphones has highlighted a slight difference between the original and the processed speech signal. However, especially for noisy speech, this difference does not degrade the perceived speech quality. The requirement of MOS $\geq$ 3.6 is always guaranteed, and in most cases MOS $\geq$ 4.1 is achieved, an excellent quality for ATC communications.

6. Conclusions

This paper considered a promising approach for secure communications in ATC systems based on an embedded watermarking method. By means of the proposed method we have *a hidden high capacity data channel* able to support a maximum data rate of about 5000 bits/s (dynamically variable thanks to segmentation algorithm parameters). Note that the achieved maximum data rate value is 20–50 times greater than that allowed in current state-of-the-art traditional speech watermarking systems. Important advantages of this method are also *a higher voice quality* (MOS $\geq$ 4.4), *high robustness with respect to channel noise*, low implementation complexity as the system requires only low-cost and easy-to-install encoders and decoders, and full compatibility with legacy RF DSB-AM transceivers. Moreover, it is well suited both for the actual 25 kHz RF channels and for the new 8.33 kHz European RF channels.

In addition to this, the supported data rate, well above that needed to support the AC signature application requirements, *makes the proposed system suitable to support high speed data services or provide specific information as AC position and speed, sensors telemetry, or navigation parameters*. We are studying the very interesting idea of treating this hidden data channel as a packet-based digital link where we can multiplex various services with different priority classes.

As concluding remark, the practical feasibility of the proposed method has been positively tested by carrying out its implementation on a Spectrum Digital TMS320C6416 DSP Starter Kit (DSK) [31,32].

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